



A Hybrid Model for Stock Market Prediction with Stacked Autoencoder for Feature Engineering

Maryam Sadat Hashemipour¹ · Morteza Zahedi¹ · Mansoor Fateh¹

Received: 11 January 2025 / Accepted: 4 January 2026
© The Author(s), under exclusive licence to Shiraz University 2026

Abstract

Recently, the use of deep learning network concepts in the field of financial time series has attracted the attention of many researchers due to the noise and non-stationary characteristics of these data. This study presents a new deep learning framework in which the appropriate features of Tehran company's stock data are extracted using a stack auto-encoder and based on the information of the past 20 days of each stock, its future 10-day status is predicted. In first step, the generated dataset sent to a variable autoencoder to remove noise. In second step, a convolutional autoencoder is used to efficiently generate high-level features. In third step, high-level noise free features are fed into the hybrid model based on Extreme Gradient Boosting (XGBoost), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM) to forecast the next ten day price labeling. The results of this research have been evaluated in four different case studies. When the Hybrid 4-Classifer Stacked Autoencoder (H4CSAE) model was used to reduce noise and extract features, the precision is equal to 77.78%, while in other methods such as XGboost, LSTM, GRU and BiLSTM are respectively equal to 52.38%, 47.37%, 63.33% and 65.63%. Based on the results of the research, one of the strengths of the proposed method is reducing the rate of false positives in the final prediction.

Keywords Stock price prediction · Stacked autoencoder · Hybrid model · XGBoost · BiLSTM

1 Introduction

Prediction in today's world is a challenging part of human life. Hope for a better life has led man to want to better control the events ahead by predicting future events and providing the ground for improving future events as much as possible, or preventing possible losses by estimating harmful events. Among various issues, time series prediction is very important. One of the most widespread and difficult problems in time series is predicting the trend of the stock market because it is essentially dynamic, nonlinear, complex, and noisy. Numerous factors affect stock markets including economic variables, industry-specific variables, exchange rates, psychological variables of investors and political variables (Enke and Thawornwong 2005; Wang et al. 2011). Because of the noise and volatility features of

stocks, stock selection is an important but difficult topic in financial market research and analysis (Mostafa 2010). In stock market technical analysis focuses on financial time series data and is more useful in short-term transactions. Technical indicators are important parameters that are calculated from stock time series data and their purpose is to predict the direction of the financial market. In technical analysis what stock market investors are advised by consultants is not to cite the results of a graph analysis index for analysis and decision-making, and to emphasize the need to use several analysis indicators simultaneously to confirm the result. Because relying on the information of an indicator increases the probability of failure and losses.

Another important issue is to pay attention to the nature of financial data. In financial data, what happened in the past may be repeated in the future. Deep learning algorithms can detect hidden and dynamic patterns in data through their learning process. Time series predictions should be made using a deep nonlinear topology, according to recent research in pattern recognition and machine learning. Complex real world can be modelled based on deep learning networks because an important aspect of deep models is

✉ Morteza Zahedi
zahedi@shahroodut.ac.ir

¹ Faculty of Computer Engineering, Shahrood University of Technology, Shahrood, Iran

extracting strong feature sets from raw data and use them to make predictions (Bao et al. 2017; Hoseinzade and Haratizadeh 2019).

In this paper, an approach using a stacked autoencoder deep network for noise elimination and feature selection and an XGBoost binary classifier as the advanced implementation of a gradient boosting algorithm (Spoorthy and Sanjeevi 2023; Xia et al. 2017; Zięba et al. 2016) and BiLSTM model as a Recurrent Neural Network (RNN) based method as a model to retain past information to predict feature event (Siami-Namini et al. 2019) is introduced. The main contributions of this article can be summarized as follow:

1. This research introduces a state-of-the-art labeling method in terms of selecting the stock company shares.
2. This study improves the accuracy of the proposed method by removing noise and extracting data features with the assistance of the deep stack autoencoder, taking into account the capacity of deep learning-based models to extract a set of strong features from raw data.
3. Combining the results of various classifications methods XGBoost, LSTM, BiLSTM, and GRU for selecting profitable stocks. In similar studies, this technique is not very common, because most of them are mainly based on price forecasting. Decision-making based on the results of the four models reduces the level of risk associated with stock selection.
4. Stock selection is based on combining the results of the XGBoost with optimized hyperparameters due to its capability of handling regularization and overfitting-underfitting issues and the BiLSTM method due to its ability to learn long-term dependencies from past data.

The structure of this paper is as follows: In Sect. 2, related works are summarized. Section 3 is related to the introduction of the dataset and the proposed hybrid model. Section 4 analyzes the results of different case studies. Section 5 summarizes the paper's content, conclusions and future works.

2 Related Works

Machine learning models like Artificial Neural Networks (ANNs) (Guo et al. 2014) and Support Vector Regression (SVR) (Huang et al. 2005), have been widely used in research over the past few decades to accurately predict financial time series (Lu et al. 2009). Customary statistical techniques like linear regression, Auto Regression and Moving Average (ARMA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) were of great interest due to the interpretability of their results for prediction financial time series. However, given the complexity

of financial time series, one of the most appealing topics is combining deep learning with market prediction (Bao et al. 2017).

Bachelier (1900) in 1900 was the first to randomly determine stock price changes and was one of the pioneers of studies on stock price behavior. The idea of Malkiel and Fama (Fama 1970) They found that past and present information have no effect on future stock prices in their research. They presented the efficient market hypothesis. Parametric statistical models like ARMA, Auto Regressive Integrated Moving Average (ARIMA), and Vector Autoregressive (VAR) are utilized in conventional methods for making predictions about time series and locating the most accurate estimates (Box et al. 2015; Virtanen and Yli-Olli 1987).

The ARIMA model is useful for evaluating the relationships between variables through statistical inference, according to the findings of (Ariyo et al. 2014), but it is not suitable for financial time series analysis. Because the ARIMA model is linear and it is not able to detect the non-linear nature of stock prices. Financial time series data are extremely noisy and exhibit time-varying fluctuations, ARIMA approach is predicated on the assumption that the variance of the data is constant.

Due to the limitations of statistical methods, researchers made significant efforts to model nonlinear relationships in financial time series using machine learning techniques (Zare Mehrjardi et al. 2016). Because of their capacity for generalization and nonlinear mapping, artificial neural networks and support vector machines have been taken into consideration. ANN, in contrast to statistical models, does not necessitate a set of imposed hypotheses and will work if there is sufficient data (Hornik et al. 1989). In addition to machine learning, some studies (Kim and Shin 2007; Savargiv et al. 2024; Yang et al. 2019) use computational intelligence algorithms to discover the future characteristics of stock markets due to their powerful learning ability and high predictive accuracy for stock prices and stock returns.

In (Nobre and Neves 2019), multi-objective optimization genetic algorithm helps to improve the results of a XGBoost binary classifier. In this research, PCA and Discrete Wavelet Transform (DWT) methods are combined to reduce the dimensions and noise elimination. The use of PCA and DWT as the stage of pre-processing of the data set and the optimization of the hyperparameters of the XGBoost method has led to the design of a model that has obtained acceptable results in the financial markets. The analysis of the results shows that the designed system, due to the use of Harr wavelet transform, has well recognized the sudden changes of a financial signal and has provided a very good performance in periods of high-oscillating.

Chou and Nguyen (2018) make use of sliding window meta-optimization to create a smart and efficient system to

predicted time series. In this research predicting construction company stock prices by incorporating a hybrid model of a Metaheuristic Firefly and Least Squares Vector Regression (MetaFA-LSSVR) algorithm is done. MetaFA has been selected as the training algorithm to increase LSSVR learner performance by automatically adjusting the C and σ hyperparameters of the Radial Basis Function in LSSVR. The proposed model was tested in short-term and long-term forecasts, furthermore the findings demonstrate that as the number of forecasted days increases, so does the model error.

Recently, deep learning approaches have been considered by researchers in financial forecasting problems because of their deep nonlinear structure (Jiang 2021; Mukherjee et al. 2023; Nabipour et al. 2020). Forecasting stock prices requires both current and historical information, so RNN deep return networks are well suited for time series forecasting due to their memory (Guo et al. 2016). In recent studies, the LSTM deep model, which is well-designed for learning time patterns, is widely used for time series analysis.

In (Long et al. 2019), combining two types of Convolution and LSTM networks, a deep learning-based multi-filter neural network model is suggested for extracting features in multivariate financial time series and trend forecasting. The Chinese stock market index feature results of the proposed model show that deep learning-based methods are superior to conventional statistical and machine learning methods. In market simulation, deep learning models outperformed all other baselines in terms of profitability.

The financial time series prediction capabilities of the BiLSTM, LSTM, SVR, and ARIMA models are contrasted in 2022 (Yang and Wang 2022). In this research the close price was used to evaluate the model results. The test set's application of the model with BiLSTM layers achieves the minimum Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE), as demonstrated by the results. This study demonstrates that when processing financial time series, machine learning models that are able to comprehend and analyze nonlinear relationships in close price data perform better than linear models like ARIMA methods. In addition to these findings, it was demonstrated that the BiLSTM model can successfully predict noisy financial time series.

In article (Ahangari and Sebt 2023), a hybrid method is introduced that uses a lexicon and SVM algorithm to present a new model for sentiment analysis of Iranian Stock Market. The feature vector is calculated from the constructed lexicon made by using stock price information related to user's opinions. In order to categorize the comments of each user, 3-class classification with three classes: positive, negative, and neutral was used. The conducted tests show that the

use of the proposed method has an improvement in F-Score compared to the baseline method.

In paper (Spoorthy and Sanjeevi 2023), a hybrid model of multilayer BiLSTM and Bayesian Optimization LightGBM based on the Pearson correlation coefficient is proposed to solve the problem of stock price forecasting. The training and the testing set are made by stock attributes "Open", "High", "Low", "Close" and "AdjClose". The multilayer bidirectional LSTM model is used to train the attributes and prediction results are used to build a new testing set. Finally, the BO-LightGBM model is used to predict the new testing set. This model has better approximation and generalization ability in stock price forecasting.

To model stock price prediction via the Generative Adversarial Network, which is based on the Generator and Discriminator combination, authors have included TSGAN as the main subject. Real data is analyzed to evaluate the advantages of the proposed TSGAN and compare it to other models. Nevertheless, the empirical results revealed that the TSGAN is the best in the stack and its proposed model's accuracy in stock prediction is much better than other methods. Instead of using a deep learning model to determine future prices and making top decisions, traders can now rely on AI that helps in identifying risk exposure (Xu et al. 2024).

The manuscript proposes MDF-DMC model, which is composed of the multi-view stock data features and dynamic market correlation information, for stock trend prediction. As the model uses MLP-Mixer to extract the stock trend features and an improved encoder to dynamically learn the correlation dynamics between the stocks and the market, the stock features become an integral part of the model. The model definitely does give us more accurate predictions or forecasts with this new integration. Our results prove that MDF-DMC tops the other state-of-the-art methods in terms of prediction accuracy, generalization ability, profit making, and risk reduction. This model's purpose is to predict the stock of a particular company for the next day from the data of that day, so the process needs to be more efficient (Yang et al. 2024).

This study employs deep learning in conjunction with feature selection and classification techniques in light of the success of deep networks in analyzing financial time series data and the impact of selecting suitable features for the input data set.

Although many other hybrid deep learning architectures, including Transformer-based structures or GAN-based structures, have emerged lately for financial forecasting tasks, these structures are mostly aimed at modeling the sequence dependencies or the generative reconstruction tasks, respectively. However, the proposed solution combines multi-view feature learning with dynamic correlation

Table 1 Comparative summary of related hybrid deep learning models for financial forecasting

Study	Year	Model/framework	Data source	Main contribution	Limitation
(Bao et al. 2017)	2017	LSTM+Wavelet+Autoencoder	Shanghai Stock Exchange	Early hybrid model combining wavelet transform and stacked autoencoder for feature extraction	Lacks ensemble classifiers; limited robustness to noise
(Nabipour et al. 2020)	2020	CNN–LSTM Hybrid	Tehran Stock Exchange	Used CNN for local pattern extraction and LSTM for temporal forecasting	No denoising or feature selection stage
(Long et al. 2019)	2019	Conv-LSTM	Chinese stock index	Multi-filter convolutional layers capture spatial–temporal features	Overfitting and low interpretability
(Spoorthy and Sanjeevi 2023)	2023	BiLSTM+BO-LightGBM	Indian Market	Uses bidirectional LSTM for feature generation and LightGBM for prediction	Feature redundancy and no noise-handling mechanism
(Xu et al. 2024)	2024	TSGAN (Time-Series GAN)	Real-world Stock Data	GAN-based sequence modeling for synthetic data augmentation	Training instability and mode collapse risk
(Yang et al. 2024)	2024	MDF-DMC (MLP-Mixer+Dynamic Correlation Encoder)	Chinese A-shares	Incorporates multi-view features and dynamic market correlations	High computational cost; short-term prediction only
Proposed H4CSAE (This Study)	2025	Stacked VAE+CAE+XGBoost+LSTM+GRU+BiLSTM	Tehran Stock Exchange	Combines stacked VAE–CAE for denoising and feature extraction with four ensemble classifiers to reduce false positives and improve precision	Tested on one market; no sentiment or macroeconomic features

Table 2 Comparative analyses of architectural approaches

Model type	Key strengths	Limitations	Data requirements
H4CSAE (Ours)	Feature denoising, Multi-perspective learning	Ensemble complexity	Medium
Transformer-based	Long-range dependencies, Attention mechanisms	High computational cost	Large
GAN-enhanced	Data augmentation, Robustness	Training instability	Large
Traditional Hybrid	Interpretability, Stability	Limited feature learning	Small

learning, which makes the solution different from other existing autoencoder-based solutions, including the solution proposed by (Bao et al. 2017), due to the novel correlation learning mechanism integrated into the deep learning solution. Table 1 is a comparison of the recent deep learning techniques developed for financial stock price prediction. Although the current state-of-the-art techniques have brought about numerous innovations in their design, some important aspects, including noise reduction, feature selection, are largely ignored in these techniques. The proposed H4CSAE framework fills the gap by efficiently utilizing the capabilities of stacked autoencoders with an ensemble of different classifiers which collectively improve predictive precision and reduce false positives.

Besides comparing our work with existing hybrid methods, we should also evaluate it against newer architectures. Transformer models like Informer (Zhou et al. 2021) and TFT (Zhou et al. 2021) understand data patterns well but need strong computers and lots of data to work properly. Similarly, GAN-enhanced models (Wiese et al. 2020; Yi et al. 2025) are good at improving data but can have issues with staying stable during training. The H4CSAE framework has a different balance to consider.

The H4CSAE framework involves a different trade-off. While it does not target ultra-long sequences like Transformers, it offers a computationally efficient and robust precision-focused forecasting mechanism for noisy, medium-length financial series—a common scenario that one may face in the real world. It does this through stacked feature denoising and a conservative ensemble to avoid instability from GANs. Hence, H4CSAE is not an incremental improvement but a pragmatic alternative tailored for reliable financial decision-making. A high-level comparison of these architectural paradigms is summarized in Table 2.

3 The Proposed System

3.1 Classifiers

In past research (Ravi et al. 2017), to solve the challenging problem of financial time series prediction, researchers have concluded that the appropriate combination of different forecasting methods can provide better results than stand-alone methods. A good hybrid forecasting model can reduce the uncertainty of the model in addition to improving the

Table 3 Introduction of four used classifiers

Classifier	Efficiency	Parameter
(Chen and Guestrin 2016)	Ability to perform parallel calculations	max_depth (Budholiya et al. 2020)
	Favorable performance in reducing overfitting due to proper generalizability (Jiang et al. 2020)	n_estimators
	Higher speed compared to similar classification methods	learning_rate (Budholiya et al. 2020)
	Creating an acceptable balance between the two aspects of bias and variance	
Long Short Term Memory (LSTM) (Hochreiter and Schmidhuber 1997)	Passes the important information along the sequence chain to get the desired output (Apaydin et al. 2020; Papanicolaou et al. 2023)	
	Good way to deal with vanishing gradients (Bao et al. 2017)	
Gated Recurrent Unit (GRU) (Cho, Van Merriënboer, Bahdanau, & Bengio, 2014)	A modified version of LSTM (Mateus et al. 2021)	
	Using less memory than LSTM (Apaydin et al. 2020)	
Bidirectional LSTM (BiLSTM) (Graves and Schmidhuber 2005)	Twice for training, which results in improving the learning of long-term dependencies and increasing the accuracy of the model (Siarni-Namini et al. 2019; Yang and Wang 2022)	

Table 4 List of the 20 features output to the data pre-processing

Technical Indicators	# Features	Time period
Relative Strength Index (RSI) (Caldera and Lavanya 2020; Shynkevich et al. 2017)	1	14 days
Moving Average Convergence Divergence (MACD) (Basak et al. 2019)	3	26 and 12 days for MACD line- 9 days for Signal Line
Stochastic RSI	2	3 days
Bollinger Band (Shynkevich et al. 2017)	4	20-day
Average Directional Movement Index (ADX) (Gurrib 2018)	1	Default value in reference
Ichimoku (Deng and Sakurai 2014; Patel 2010)	5	Default value in reference
Chaikin Money Flow Indicator (CMF) (Caldera and Lavanya 2020)	1	20–21 days
Raw financial data	# Features	
Close	1	
Open	1	
Volume	1	

forecasting performance. A financial time series is highly uncertain, noisy and incomplete under the influence of economic, social and political factors, so we need to develop hybrid techniques to obtain more efficient prediction results (Savargiv et al. 2024). According to these cases, the proposed model of this research predicts financial data based on the results of four classifiers which introduced in Table 3.

3.2 Technical Analysis Module

Due to the absence of a properly organized dataset containing financial news for the Tehran Stock Exchange, this study takes into account historical stock prices and technical indicators only. The input of the technical analysis module is raw financial data, and the output is selected technical indicators that are utilized for stock prediction. To learn effectively, we employ multiple indicators rather than a single one since each captures various dimensions of historical market action

and investor preferences (Caldera and Lavanya 2020; Nobre and Neves 2019).

The selected 17 technical indicators, in addition to three raw features (Open, Close, and Volume), work over varied time frames (e.g., 14-day, 26-day) and employ varying price points (high, low, close) and volume, allowing the model to learn to identify various patterns in the data. This provides a dataset of 20 features, calculated through MATLAB software (see Table 4).

3.3 Label Generation Module

After deriving 20 technical features, we applied a 20-day sliding window during data processing of financial time series data from Tehran Stock Exchange. The goal is to forecast whether each and every stock will realize a profit greater than 17% over the following 10 days. This threshold value is chosen to have adequate positive samples to train and to reflect reasonable investor expectations: smaller

thresholds will include stocks with marginal profitability and have less useful significance, while larger thresholds will have very sparse positive samples and induce high class imbalance with less reliable models. Those stocks above this threshold are labeled with 1 (profitable) and others as 0 (non-profitable). The sliding window moves with and generates labeled samples for training set by utilizing difference of 30th day (prediction period end) and 20th day (sliding window end) close prices. This process generates a balanced and realistic labelling scheme for efficient stock picking.

3.3.1 Empirical Validation of Threshold Selection

We performed an empirical analysis of return distributions across our dataset ($N=535$ companies) to supplement the conceptual justification. The findings show that 67.7% of stocks yield returns below 17% (marginal-to-moderate profits), while 42.5% achieve returns $\geq 17\%$ (meaningful profits), suggesting that the 17% threshold is the best way to distinguish meaningful profit opportunities from market noise. This distribution demonstrates that the 17% threshold preserves an ideal balance by eliminating noisy marginal returns while maintaining enough positive samples (42.5%) for reliable model training. Higher thresholds would result in an unacceptable class disparity, while lower thresholds would introduce excessive noise.

According to Eq. (1), if the 30th day closing price (the tenth and final day of the period) is higher than the 20th day (the last day of the sliding window), it means that the stock will have an upward trend in the future. Equation (2) will be calculated.

$$MaxPrice = maximum(close(21 : 30)) \quad (1)$$

$$labelTure = \left(\frac{MaxPrice - close(20)}{close(20)} \right) \times 100 \quad (2)$$

According to Eq. (3), if the 30th day closing price (the tenth and final day of the period) is less than the 20th day (the last day of the sliding window), it means that the stock will have a downward trend in the future, so the label of this window with Eq. (4) is obtained:

$$MinPrice = minimum(close(21 : 30)) \quad (3)$$

$$labelTrue = \left(\frac{MinPrice - close(20)}{close(20)} \right) \times 100 \quad (4)$$

In this way, the entire data of the training data set is scrolled and labelled. Finally, we will divide the data set classes into two classes by Eq. (5).

$$label(t) = \begin{cases} +1 & labelTure(t) \geq 17 \\ 0 & otherwise \end{cases} \quad (5)$$

3.4 Data Preparation Module

After constructing the features, we divided the dataset into a train set and a test set, with the most recent 10 trading days for each company forming the test set and the remaining days forming the train set. In order to mitigate class imbalance and reduce the rate of false positives (FP), directly convertible into investors' financial loss, we applied a sample reduction approach. Specifically, twice the number of negative (label 0) samples were assigned than for positive (label 1) samples, which were randomly selected. Experimental findings showed that this arrangement enables the model to learn the patterns of negatives more efficiently, and hence, reduces false positives and more stable results.

We used a walk-forward validation protocol to stop data leakage in our time series forecasting model. Only data from the 2011–2020 period was used to train the model, and its testing was restricted to the next 10 days to ensure that no data from the future was included. Finally, all feature values were normalized in the interval $[0,1]$ through Min-Max normalization Eq. (6).

$$X_{normalizaion} = \left(\frac{X - X_{min}}{X_{max} - X_{min}} \right) \quad (6)$$

3.5 Stacked Autoencoder for Denoising and Feature Extraction

In this module autoencoders are applied for training the dataset features (Volume, Close and Open price, and technical indicators). The hidden layers of the network in the autoencoder structure reduce the input's dimensionality and learn relevant features for effective reconstruction. A number of non-linear representational layers are also used by deep autoencoders to learn complex hierarchical features from data that contain a lot of information. Most of the time, the ultimate target of dimensionality reduction is not just to decrease the number of dimensions but also to reduce this number while maintaining the majority of the information about the data structure in the reduced representations. Therefore, the non-linear modality of financial time series data should be considered when selecting a method.

Another difficulty with stock data is the presence of noise caused by external economic and political factors. Hence, the feature extraction method should not only eliminate such noise but also identify patterns present beneath them that may recur in future market conditions. Previous work

suggests that autoencoders are capable of addressing both noise removal and feature representation.

Two components of stacked autoencoder structure was implemented in this study are (Fig. 1):

1. A Variational Autoencoder (VAE) To Denoise the Labeled and Normalized time Series Data
2. A Convolutional Autoencoder (CAE) To Extract deep-level Features for Subsequent Classifiers

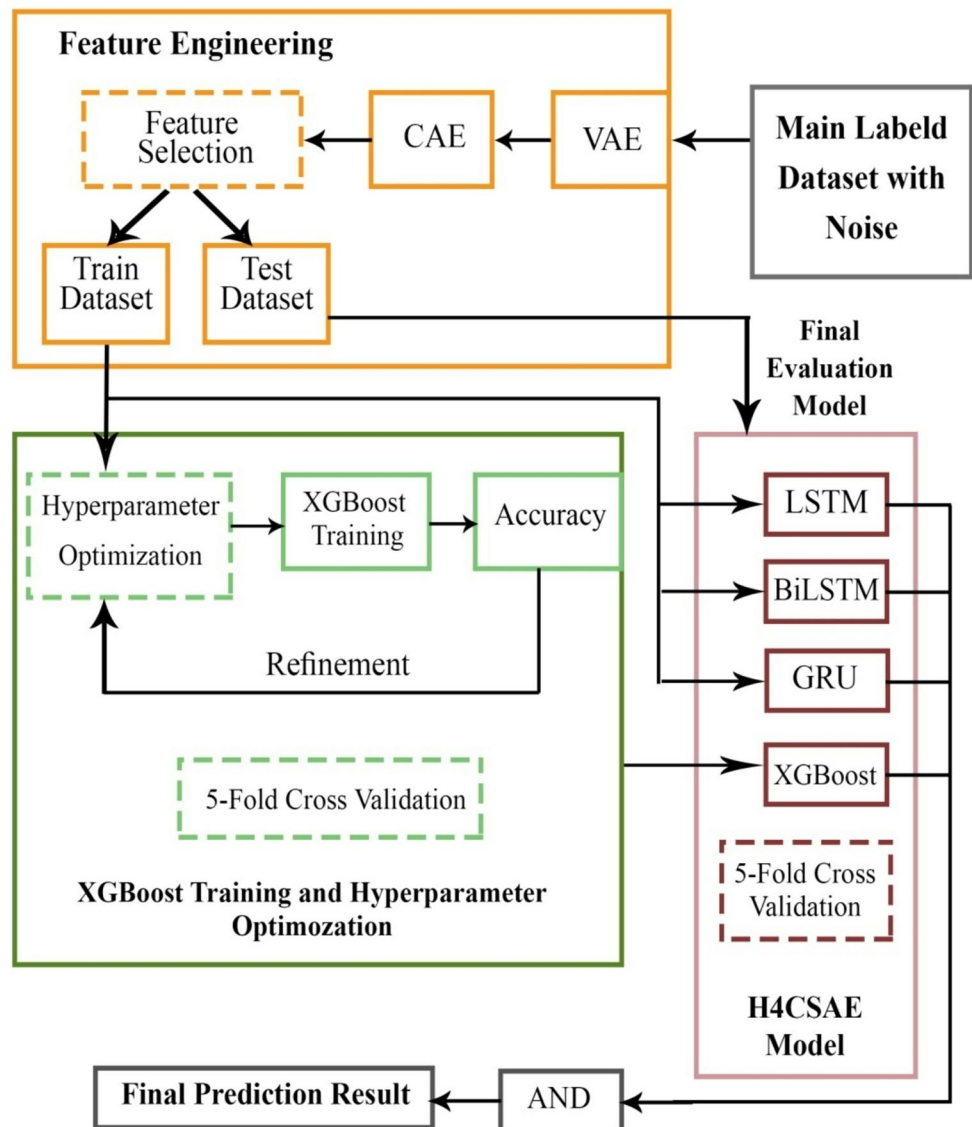
The original data consisted of 20 features that were compressed into 10 representative and denoised features in this module.

3.6 Proposed Hybrid Model

As argued by (Brownlee 2017), the selection of features has an exceedingly significant impact on the performance of the XGBoost algorithm. Feature selection is therefore necessary. In this study, hyperparameters of XGBoost are optimized using HyperOpt (Komer et al. 2019), an open-source Python library that employs a Bayesian optimization approach. The binary classifier based on XGBoost is then applied using the official Python package for XGBoost¹.

Until now, there is currently no ideal criterion for designing a neural network's optimal structure. Our experiments add a dropout layer to the model to prevent overfitting (Gal & Ghahramani 2016), In this paper the dropout rate is set

Fig. 1 Hybrid model H4CSAE for stock market prediction with stacked autoencoder for feature engineering



¹ <https://xgboost.readthedocs.io/en/latest/>.

to 0.2 same as (Yang and Wang 2022) research. Yang et al. found that a dropout rate of 0.2 is suitable for various tasks.

Kingma et al. suggest that the Adam optimizer be used for optimization training (Kingma and Ba 2014). Adam is a deep learning model training optimization algorithm that takes the place of stochastic gradient descent. By combining the best features of the AdaGrad and RMSProp algorithms, it can handle sparse gradients on noisy problems and performs well in nonlinear and unsteady problems (Khan et al. 2020; Yang and Wang 2022). It's simple execution, little memory necessities, and suitability for non-stationary make Adam a productive calculation that has been utilized effectively (Chang et al. 2019). In many research, Adam has been utilized for stock prediction the most (Ding and Qin 2020; Jain et al. 2018; Jiang 2021; Lu et al. 2021; Maggipinto et al. 2018; Wang et al. 2022; Yang and Wang 2022).

The following are the parameters that must be manually set for the model: the number of hidden layers of BiLSTM, GRU, and LSTM are one. Batch size of all classifier is set to 64, and Epoch is set to 30. Finally, binary crossentropy is the loss function for our binary classification problem. The aforementioned deep learning models are fitted using the Keras Python package. After preparation the classifier the noise-free data will be sent to four classifiers to predict the stock price trend of each company in the next ten days, and according to the subscription of these four classifiers results, the final decision to buy or sell company stocks will be made. For getting the privileged system efficiency from each experiment, three different runs were performed. The proposed method in this research works according to steps of Algorithm 1 algorithm.

Input: Raw market data (OHLC) for all Tehran Stock Exchange companies from 2011-03-16 to 2020-03-19

Output: Final portfolio of selected stocks

1. Data Preprocessing & Feature Engineering

- 1.1. Collect OHLC (Open, High, Low, Close) prices and daily trading Volume.
- 1.2. Reserve the last 20 trading days of each stock as the test set.
- 1.3. Calculate the technical indicators listed in Table 4.
- 1.4. Add the calculated indicators as features to the dataset.
- 1.5. Label the dataset for the forecasting task.

2. Feature Denoising and Reduction

- 2.1. Normalize the entire dataset.
- 2.2. Pass the normalized data through a Variational Autoencoder (VAE) to remove noise.
- 2.3. Feed the denoised data into a Convolutional Autoencoder (CAE) for high-level feature extraction.
- 2.4. Obtain the final set of refined features for the training dataset.

3. Model Optimization & Training

- 3.1. Optimize hyperparameters for all base classifiers:
 - XGBoost: Bayesian optimization with 5-fold cross-validation
 - LSTM/GRU/BiLSTM: Grid search with early stopping
- 3.2. Train the four optimized classifiers (XGBoost, LSTM, GRU, BiLSTM) on the refined training set.

4. Ensemble Prediction

- 4.1. For each company, obtain binary predictions (Buy/Hold) from all four models.
- 4.2. Apply the AND-logic rule: Final Prediction = XGBoost $\wedge$ LSTM $\wedge$ GRU $\wedge$ BiLSTM

5. Final Stock Selection

- 5.1. Execute the entire pipeline (Steps 1-4) three independent times.
 - 5.2. Select stocks that are consistently recommended across multiple runs.
 - 5.3. Generate the final portfolio.
-

Algorithm 1 H4CSAE method framework

Table 5 Dimensionality flow of the proposed H4CSAE framework, showing the input and output shapes at each feature processing stage

Stage	Input dimension	Output dimension	Transformation description
Raw Data	(n, 20, 20)	(n, 20, 20, 1)	Reshaping for convolutional processing
VAE Encoder	(n, 20, 20, 1)	(n, 2)	Dimensionality reduction to a latent space
VAE Decoder	(n, 2)	(n, 20, 20, 1)	Data reconstruction
CNN Encoder	(n, 20, 20, 1)	(n, 20, 10, 1)	High-level feature extraction
Feature Split	(n, 20, 10, 1)	(n, 100) & (n, 10, 10)	Reshaping for tree-based and sequence-based classifiers
Ensemble Output	Multiple inputs	(n, 1)	Final binary prediction via AND logic

Our framework combines the ensembles using a logical AND, which is in line with our design choice to reduce false positives in financial applications. The final prediction is generated as follows:

$$\hat{y}_{final} = \hat{y}_{XGBoost} \wedge \hat{y}_{BiLSTM} \wedge \hat{y}_{LSTM} \wedge \hat{y}_{GRU} \quad (7)$$

Equation (7) shows the binary (0/1) prediction produced by each base classifier. This conservative approach ensures that the “buy” signal is only activated when all four models reach a single consensus. As shown in the Results section, the AND combination of the 4-class output significantly increases the prediction precision and reduces false positives. The input and output dimensions at each stage of the proposed H4CSAE model processing are specified in Table 5. The variable autoencoder (VAE) facilitates denoising and nonlinear dimensionality reduction into a 2D hidden space, while the convolutional autoencoder (CAE) subsequently extracts high-level features from this cleaned data.

4 Evaluation

The model was trained and tested on Tehran Stock Exchange data (535 companies, 2011–2020). Although the H4CSAE framework is validated in this market, it is generalizable and can potentially be applied to other markets with similar structural characteristics. The H4CSAE model was trained only on Tehran Stock Exchange data. Because

of this, possibility of overfitting is a concern. To address this concern, we conducted cross-validation for hyperparameter tuning, XGBoost regularization, and early stopping in the recurrent models (LSTM, GRU, BiLSTM). While overfitting cannot be completely eliminated, these method help that the model focus on learning generalizable pattern instead of memorizing training data.

4.1 Evaluation Metrics

Our research’s evaluation metrics are chosen from widely used machine learning criteria that can reflect the candidate models’ overall prediction performance. The accuracy value is used to evaluate the proposed models’ performance in this study. The evaluation metrics are defined in Eq. (8) which requires estimating the number of True Positive (TP) and False Positive (FP) samples. But another point that is more important in this research is reducing the FP rate. Because based on the assumptions of this research, a profitable investment needs to choose stocks of companies that guarantee more than 17% profit in the next ten days.

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

4.2 Experimental Results

Case Study I - Influence of Removing all Autoencoder Types The first case study examines the performance of

Table 6 Experimental results of **Case study I**: model performance on the Raw Tehran stock exchange dataset (2011–2020) without autoencoder-based denoising or feature extraction

# Run		XGBoost	LSTM	GRU	BiLSTM	H4CSAE
1	TP	4	75	53	111	1
	FN	223	152	174	116	226
	FP	4	89	60	141	1
	TN	304	219	248	167	307
2	TP	6	25	20	137	0
	FN	221	202	207	90	227
	FP	6	29	20	197	0
	TN	302	279	288	129	308
3	TP	0	19	52	31	0
	FN	227	208	175	196	227
	FP	0	14	55	43	0
	TN	308	294	253	265	308

the proposed system when the autoencoder methods are completely excluded. To this purpose, the baseline system was evaluated over the raw data without performing denoising or feature extraction by using the autoencoder-based methods. The results of this experiment are summarized in Table 6. According to the actual trading outcomes, 227 companies achieved profits exceeding 17% within the next ten days, while 308 companies reported profits below this threshold. The analysis indicates that autoencoder methods (both feature extraction and noise removal) play a vital role in improving the system's performance. Their absence not only diminishes the model's capability to accurately classify successful companies but also significantly increases the false positive (FP) rate.

As illustrated Table 6, nearly all three implementations of the presented approach (except for the third one based on LSTM), the FP rate is greater than the true positive (TP) rate across all deep learning models. When evaluating the combined performance of the four classifiers, the issue of imbalance between positive and negative growth rates becomes even more pronounced. In the second and third implementations, no stocks were identified as profitable, while the first implementation managed to add only two stocks. However, one of these stocks (Iran Ind, with a growth rate of -4.5%) ultimately resulted in a financial loss for the investor.

The results indicate that removing autoencoder-based processing provides low performance in stock prediction. Additionally, the importance of designing accurate features for the XGBoost classifier is underscored. To optimizing the number of TP samples and FP, both noise elimination and dimensionality reduction must be implemented. As the hybrid system relies on combination results of four classifiers, the failure of XGBoost in feature usage undermines the entire system's ability to recommend value stocks.

Case Study II - Influence of Using convolution-base Autoencoder for Feature Selection In this case study, according to the results of the first case study to improve the generalization

capability and robustness of the proposed hybrid model for enhancing the quality of selected features and better learning the non-linear relationship between selected features a convolution-based autoencoder model is developed.

For this case study, no effort was made to remove the random noise in the data which is attributed to political, economic, and social fluctuations. According to the information in Table 7 at this stage, the classifiers introduced in Table 3 are not able to correctly classify the samples, which results in an increased FP rate for the hybrid model. Due to the importance of reducing the FP rate, it is necessary to use a method to smooth and remove random noises before creating suitable features in the data set. Noisy data causes bad feature and finally, lead to worse generalization and predictive power of both the tree-based and deep learning classifiers. According to the results of Table 7, the classifier based on GRU has provided a more inappropriate result.

It is clear that if the process of creating features is done from unsmoothed and noisy data, the stage of feature selection with CAE will be not only beneficial but also harmful. Because the features created due to the presence of noise have deviations and greatly reduce the generalization of the model. At this stage, the importance of using noise removal and data smoothing before selecting features can be confirmed.

Case Study III - Influence of Using the PCA Method Instead of the CAE Method in Selecting the Appropriate Features In the course of the experiment, after the application of variational autoencoder (VAE) for denoising and smoothing the data set, Principal Component Analysis (PCA) method was applied to the feature selection of 10 features. Although PCA is a traditional dimensionality reduction algorithm, its linearity makes it an unsuitable method for extracting nonlinear patterns from stock market time series data. Naturally, classifiers do not do a good job at this stage. The inability of PCA to obtain effective and representative

Table 7 Experimental results of Case study II using a Convolutional autoencoder (CAE) for feature selection without prior denoising on the Tehran stock exchange dataset (2011–2020)

# Run		XGBoost	LSTM	GRU	BiLSTM	H4CSAE
1	TP	161	216	183	512	142
	FN	66	11	44	15	85
	FP	223	275	236	268	178
	TN	85	33	72	40	130
2	TP	175	215	215	202	160
	FN	52	12	12	25	67
	FP	220	279	286	264	202
	TN	88	29	22	44	106
3	TP	174	221	219	192	154
	FN	53	6	8	35	73
	FP	227	288	291	254	200
	TN	81	20	17	54	108

Table 8 Experimental results of **Case study III**: model performance using **Principal component analysis (PCA) for feature selection** (after VAE denoising) on the Tehran stock exchange dataset (2011–2020)

# Run		XGBoost	LSTM	GRU	BiLSTM	H4CSAE
1	TP	0	0	0	0	0
	FN	227	227	227	227	227
	FP	0	0	0	0	0
	TN	308	308	308	308	308
2	TP	0	4	0	0	0
	FN	227	223	227	227	227
	FP	0	2	0	0	0
	TN	308	306	308	308	308
3	TP	0	5	6	2	0
	FN	227	222	221	225	227
	FP	0	8	5	1	0
	TN	308	300	303	307	308

Table 9 Experimental results of **Case study IV**: performance of the proposed **H4CSAE framework using a stacked autoencoder (VAE+CAE) for joint noise elimination and feature selection** on the Tehran stock exchange dataset (2011–2020)

	XGBoost	LSTM	GRU	BiLSTM	H4CSAE
TP	22	9	19	21	14
FN	205	207	208	206	213
FP	20	10	11	11	4
TN	288	298	297	297	304
Precision (%)	52.38	47.37	63.33	65.63	77.78
Accuracy (%)	57.94	58.59	59.07	59.44	59.44
Specificity (%)	93.5	96.8	96.4	96.4	98.7

Bold values indicate the best performance for each evaluation metric

features—particularly for the XGBoost classifier—leaves room for very poor predictive outcomes. As can be deduced from the results shown in Table 8, in the first iteration, the classifiers were unable to predict companies with profit rates above 17% within the next ten days.

These findings reflect the confines of linear feature extraction methods such as PCA within the process of prediction. In contrast, non-linear methods such as convolutional or stacked autoencoders produce better representations and significantly improve the predictability of the hybrid system.

Case Study IV - Influence of Using Stacked Autoencoder for Feature Selection and Noise Elimination A system that not only reduces the financial input data set's dimensionality but also reduces the data set's noise has been created by combining the variational autoencoder method with the convolutional autoencoder method. This is done in order to determine whether these two techniques applied together as a stack autoencoder model leads to the improvement of the system results compared to the study cases I and II. This is because the financial input data is reduced in features by the CAE and denoised by the VAE. Having fewer features

in the dataset helps to avoid overfitting the training data and gets rid of some irrelevant samples that can hurt the performance of the system. This helps the system learn and become better at generalizing. The results of this stage of the tests are shown in Table 8. It should be noted that these results include the result of the best implementation of the 3 implementations of the introduced method in case study IV. According to the presented results, the hybrid method has best precision. Note that recall and F1-score values are relatively low due to the inherent class skewing and the priority of minimizing false positives in order to have effective practical stock picking. Precision is therefore preferred to recall and F1-score to ensure reliability and reduce investor risk. As shown in Table 9, H4CSAE achieves superior precision (77.78%) while maintaining high specificity (98.7%), demonstrating its exceptional capability to minimize false positives. This precision-focused approach aligns with financial applications where false positives lead to direct monetary losses.

The low recall in H4CSAE is an intended consequence of its conservative AND-logic ensemble design: It explicitly opts to avoid false positives at the cost of overlooking some potential opportunities. For financial applications, this precision-oriented strategy is well-placed, where the avoidance of huge potential losses due to wrong signals is more relevant than the maximization of profit opportunities. This trade-off aligns perfectly with risk-averse investment strategies where capital preservation is paramount.

Precision-Recall comparison for all models is shown in Fig. 2 to visually illustrate this property. As seen from this plot, H4CSAE is positioned in the high-precision region, which further justifies its optimality for financial risk management.

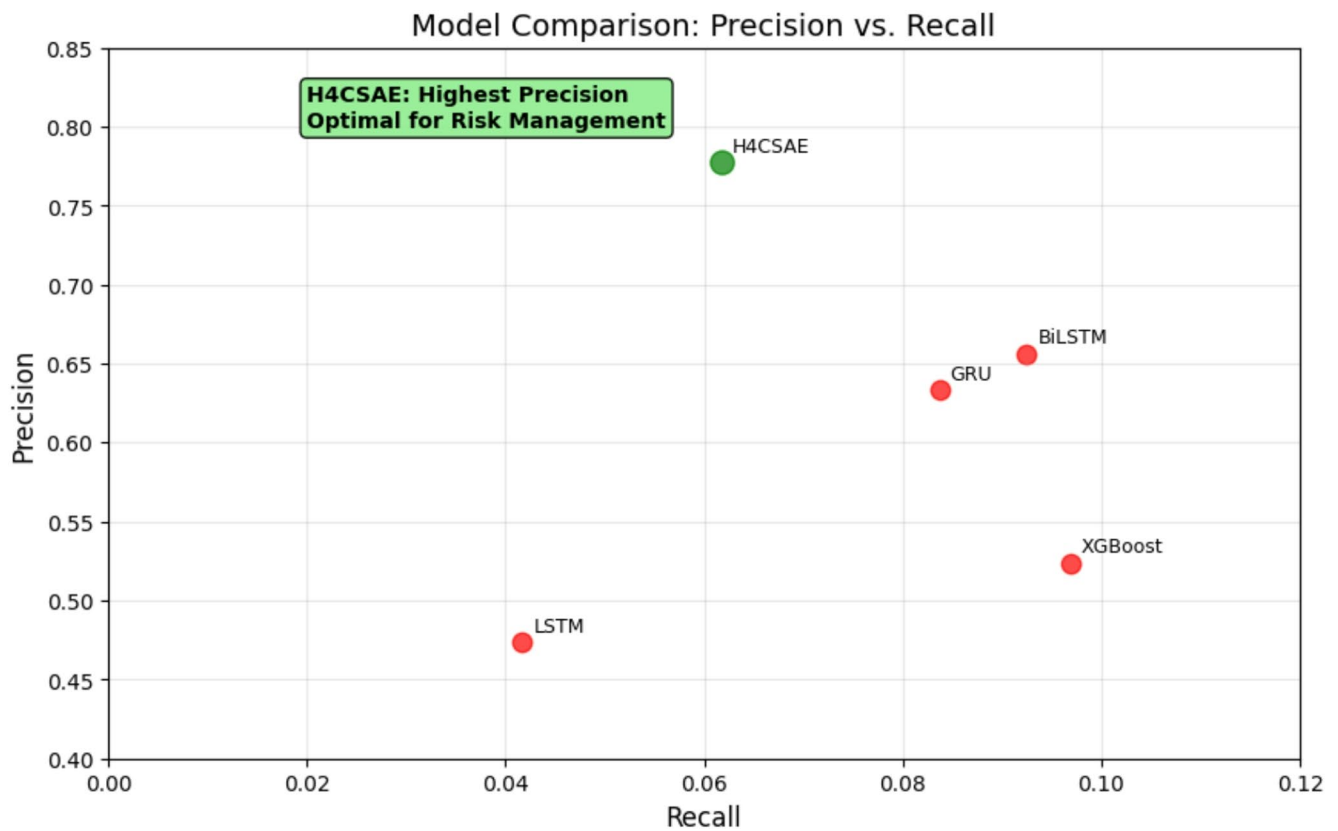


Fig. 2 Precision-Recall comparison. H4CSAE achieves the highest precision (77.78%), optimal for financial applications where minimizing false positives is critical

Table 10 McNemar's test results for **Case study IV**, comparing the classification outcomes of the proposed H4CSAE framework against each base model on the Tehran stock exchange dataset

	b (H4CSAE wrong, Base correct)	c (H4CSAE correct, Base wrong)	<i>p</i> -value (McNemar exact)
H4CSAE vs. XGBoost	0	23	<0.001
H4CSAE vs. LSTM	0	11	0.003
H4CSAE vs. GRU	0	12	0.002
H4CSAE vs. BiLSTM	0	15	<0.001

4.3 Statistical Significance and Confidence Analysis

To statistically validate the overperformance of the developed H4CSAE framework compared to individual base models (XGBoost, LSTM, GRU, BiLSTM), we performed McNemar's exact test on Case Study IV. The b/c contingency tables as shown in Table 10 are between the samples correctly classified by H4CSAE but not by the base model (b) and the samples not correctly classified by H4CSAE but correct in the base model (c).

As expected, due to the application of AND logic within H4CSAE, $b=0$ in all the comparisons. The p -values estimated indicate that H4CSAE significantly outperforms each of the separate models ($p < 0.01$). These results provide

Table 11 Precision values with 95% confidence intervals using the Wilson score method

Model	Precision (%)	95% Confidence Interval (Wilson Score)
XGBoost	52.38	(36.4% – 67.9%)
LSTM	47.37	(26.9% – 68.5%)
GRU	63.33	(46.3% – 77.7%)
BiLSTM	65.63	(48.8% – 79.3%)
H4CSAE	77.78	(54.8% – 91.0%)

Bold values indicate the best performance for each evaluation metric

statistical evidence towards the performance of the hybrid framework in eliminating false positives while being practical for stock selection.

In this study, two additional indicators “effect size” (using Cohen's method) and “95% confidence interval” (using Wilson's method) have been used in addition to the McNemar test to ensure the reliability of the comparison results. These two indicators help to determine the stability of the proposed method in addition to the degree of superiority.

- Effect Size (Cohen's h).

With the help of Eq. (9), the superiority of the H4CSAE model in precision over the base models has been measured using the Cohen index (h).

$$h = 2 [\arcsin(\sqrt{p_1}) - \arcsin(\sqrt{p_2})] \quad (9)$$

Where p_1 = precision of H4CSAE and p_2 = precision of the baseline model.

Computed effect sizes:

- H4CSAE vs. BiLSTM: $h=0.26$.
- H4CSAE vs. GRU: $h=0.31$.

These findings, while demonstrating small to moderate effect sizes, show a meaningful reduction in false positives, a critical requirement in financial prediction tasks.

- 95% Confidence Intervals (Wilson Score).

To complement the McNemar's test results, we also computed 95% confidence intervals for precision using the Wilson score method. As shown in Table 11, the interval for the H4CSAE model remains consistently higher than or equal to those of the baseline models. This pattern confirms strong performance superiority, even when using a conservative estimation approach.

4.4 Market Simulation

To evaluate the practical applicability of the proposed methodology, a simulation of the market was executed. Following existing research, the simulation assumes zero transaction costs and perfect liquidity, isolating the predictive capacity of the model. That is, the model was experimented on under ideal conditions to ensure that the results are solely a consequence of its predictive power and not of real-world market constraints such as brokerage commissions or liquidity limitations. In this manner, the simulation is a controlled environment where the impact of the predictive model on investment decision making can be genuinely evaluated.

The simulation operates based on these basic rules:

1. Buy: When the proposed model predicts the positive class (i.e., increase in stock price), system will buy 1,000,000 IRR worth of the stock at the Opening price.
2. Sell: Stocks are sold either when they yield a 17% profit or at the end of the 10-day forecast window, depending on which occurs first.

It should be noted that, according to previous research, this simulation ignores transaction costs and assumes perfect liquidity, which limits the predictive capacity of the model. However, in the real world, transaction costs and liquidity issues affect the results and could be investigated in future studies.

Table 12 Results of market simulation trades of H4CSAE model for stock market prediction with stacked autoencoder for feature engineering

Company name	Close price last test day	Selling price	Max Close price in 10 days
Tehran Stock Exch	50,418	61,281	74,487
Maskan Pardis	13,009	15,787	18,136
Iran Radiator	54,042	63,882	63,882
Oroumijeh Cem	14,675	17,307	17,307
Kh. Pegah Dairy	40,077	48,637	58,461
Sabet Khorasan	34,075	40,464	45,324
Commodity Exch	44,465	52,792	64,167
Melal Credit Inst	5993	6767	6858

Table 13 Impacts of transaction costs on portfolio returns

Transaction Cost	Net Portfolio Return	Return Reduction vs. Baseline
0.0% (Baseline)	15.5%	-
0.1%	13.7%	1.8%
0.5%	6.7%	8.8%

The profitability of the trading strategy is measured by the total return (R) as provided in Eq. (10):

$$R = \left(\frac{Portfolio_T}{Portfolio_{t_0}} - 1 \right) \times 100\% \quad (10)$$

Where $Portfolio_T$ represents the total asset value at the end of the simulation period and $Portfolio_{t_0}$ represents the initial asset value i.e. 1,000,000 IRR at the beginning of the simulation period t_0 .

In Table 12, the number of 8 companies selected by the proposed model is introduced (note that these 8 companies were selected in all three iterations of the proposed model). In this table, the sale price is equal to the price at which the stock selected will be sold after achieving the introduced sale conditions, and the maximum column is equal to the maximum price that these stocks will have during the ten-day test. To illustrate with a real example, consider the stock of Maskan Pardis (Table 12). Buying 76 stocks at the opening price of 13,009 IRR is worth 1,000,000 IRR. Selling them after 17% growth in price (15,787 IRR) provides 1,119,812 IRR, which translates to a total return of 20%.

As can be seen in Table 12, one of company (Melal Credit Inst) which failed to achieve a growth rate exceeding 17% over the course of test days, will be sold in the 10th-day. However, choosing this company does not result in financial loss for the investor, because its stock has risen 13% in the last 10 days.

We looked at the effect of trading costs on an 8-stock portfolio (Table 12) to make sure the trading strategy was practically feasible. Table 13 illustrates the net effect on returns when taking into account typical brokerage fees in the range of 0.1% to 0.5%. The strategy shows its resilience to actual market frictions by maintaining a positive return of 6.7% even in the direst scenario with a 0.5% fee. The strategy's low trading volume—just 8 trades were made during the testing period—is the reason for its resilience.

5 Conclusion and Recommendations

5.1 Summary of Contributions

This study proposed a new hybrid framework, H4CSAE, for stock market prediction by synergistically combining deep learning models and tree-based ensembles. The core of our approach is based on using a stacked autoencoder for feature denoising and extraction, then feeding these into a unique AND-logic ensemble of four classifiers, namely, XGBoost, LSTM, GRU, and BiLSTM, to minimize false positives. The framework demonstrated superior classification precision, achieving 77.8%, outperforming the leading individual models by 12–30%. Simulation trading scenarios also resulted in consistent returns between 13% and 17% on all selected stocks while reducing false positives by up to 60%, proving the practical value of this algorithm in risk-averse investing strategies. Accordingly, as supported by quantitative results Table 9 and visual analysis Fig. 2, the high-precision, low-recall profile of H4CSAE would be very valuable for more conservative trading strategies, in which reliability outweighs the quantity of trading opportunities.

5.2 Methodological Validation

Our experimental design was intended to be robust using walk-forward validation that would prevent temporal data drift. This 17% profit threshold was statistically justified, considering a balance between signal integrity-67.7% of marginal profits were excluded-and training adequacy, including 42.5% of the significant opportunities. The stacked autoencoder played an important role in preprocessing to remove noise and extract 17 high-impact technical indicators from daily data of the Tehran Stock Exchange for 2011–2020.

Concerning model optimization, hyperparameters were tuned using a grid search procedure, combining early stopping to prevent overfitting for GRU, LSTM, and BiLSTM, while a more aggressive 5-fold cross-validation scheme was conducted on the hyperparameter-reduced search space of XGBoost. The fact that H4CSAE showed consistent

outperformance across all case studies-IV, despite such different optimization strategies, reinforces the robustness and general effectiveness of our proposed hybrid framework.

5.3 Limitations and Generalizability

One of the core limitations of this research involves its focus on a single emerging market, that of the Tehran Stock Exchange. While this focus has enabled us to adapt H4CSAE to the high-noise and specific micro-structural conditions typical of such markets—which is actually a core contribution—it does limit the immediate generalizability of our results to developed markets. These are typically characterized by higher liquidity, different volatility regimes, and distinct macroeconomic factors. Hence, we position H4CSAE not as a universal solution but rather as a robust and effective framework for emerging market forecasting. Its performance on developed markets remains a very compelling avenue for future research.

5.4 Directions for Future Research

Building on this work, we propose several promising directions:

Advanced Architectures Incorporation of concepts from state-of-the-art models such as the Transformer might provide better identification of different, more complex fluctuations in stock market data.

Integration of Sentiment Analysis The market news information could reveal the influence of political and economic events on price trends, which would introduce an additional, non-quantitative dimension to the model.

Advanced Feature Engineering Future work may investigate more creative methods of noise elimination and feature selection or expand the dataset using GANs.

Class Imbalance Though the study focused on precision with the aim of reducing false positives, future studies can consider techniques for improving recall without significant loss in precision.

Integrating Explainable AI Applying techniques such as SHAP or LIME for explaining the hybrid model will be critical in generating trust in its financial predictions, which is fast becoming an expectation for financial AI research.

Real-Time Processing Construction of the model could be done to process high-frequency data, such as hourly data,

to capture intraday volatility in markets and make it more practically applicable.

Author Contributions M.Z. Conceived and designed the study. M.F. Provided suggestions for improving the deep learning model. M.S.H. Collected and analyzed the data, developed the model, conducted simulations, and wrote the initial draft of the manuscript. All authors contributed to reviewing and editing the manuscript and approved the final version for publication.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

- Ahangari M, Sebt A (2023) A hybrid approach to sentiment analysis of Iranian stock market user's opinions. *Int J Eng* 36:573–584. <https://doi.org/10.5829/ije.2023.36.03c.18>
- Apaydin H, Feizi H, Sattari MT, Colak MS, Shamshirband S, Chau K-W (2020) Comparative analysis of recurrent neural network architectures for reservoir inflow forecasting. *Water* 12:1500. <http://doi.org/10.3390/w12051500>
- Ariyo AA, Adewumi AO, Ayo CK (2014) Stock price prediction using the ARIMA model (eds) Paper presented at the 2014 UKSim-AMSS 16th international conference on computer modelling and simulation. <https://doi.org/10.1109/UKSim.2014.67>
- Bachelier L (1900) *Théorie de la spéculation* (ed) Paper presented at the Annales scientifiques de l'École normale supérieure. <https://doi.org/10.24033/asens.476>
- Bao W, Yue J, Rao Y (2017) A deep learning framework for financial time series using stacked autoencoders and long-short term memory. *PLoS ONE* 12:e0180944. <https://doi.org/10.1371/journal.pone.0180944>
- Basak S, Kar S, Saha S, Khaidem L, Dey SR (2019) Predicting the direction of stock market prices using tree-based classifiers. *North Am J Econ Finance* 47:552–567. <https://doi.org/10.1016/j.najef.2018.06.013>
- Box GE, Jenkins GM, Reinsel GC, Ljung GM (2015) *Time series analysis: forecasting and control*. Wiley
- Brownlee J (2017) *Introduction to time series forecasting with python: how to prepare data and develop models to predict the future*. Machine Learning Mastery
- Budholiya K, Shrivastava SK, Sharma V (2020) An optimized XGBoost based diagnostic system for effective prediction of heart disease. *J King Saud University-Computer Inform Sci*. <http://doi.org/10.1016/j.jksuci.2020.10.013>
- Caldera H, Lavanya W (2020) Combinatorial Impact of Technical Indicators on Price Prediction in Colombo Stock Market (eds) Paper presented at the 2020 20th International Conference on Advances in ICT for Emerging Regions (ICTer). <https://doi.org/10.1109/ICTer51097.2020.9325500>
- Chang Z, Zhang Y, Chen W (2019) Electricity price prediction based on hybrid model of Adam optimized LSTM neural network and wavelet transform. *Energy* 187:115804. <https://doi.org/10.1016/j.energy.2019.07.134>
- Chen T, Guestrin C (2016) Xgboost (eds) A scalable tree boosting system. Paper presented at the Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. <https://doi.org/10.1145/2939672.2939785>
- Cho K, Van Merriënboer B, Bahdanau D, Bengio Y (2014) Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), 1724–1734. <https://doi.org/10.3115/v1/D14-1179>
- Chou J-S, Nguyen T-K (2018) Forward forecast of stock price using sliding-window metaheuristic-optimized machine-learning regression. *IEEE Trans Industr Inf* 14:3132–3142. <https://doi.org/10.1109/TII.2018.2794389>
- Deng S, Sakurai A (2014) Short-term foreign exchange rate trading based on the support/resistance level of Ichimoku Kinkohyo (eds) International Conference on Information Science, Electronics and Electrical Engineering. DOI: 10.1109/InfoSEE.2014.6948127
- Ding G, Qin L (2020) Study on the prediction of stock price based on the associated network model of LSTM. *Int J Mach Learn Cybernet* 11:1307–1317. <https://doi.org/10.1007/s13042-019-01041-1>
- Enke D, Thawornwong S (2005) The use of data mining and neural networks for forecasting stock market returns. *Expert Syst Appl* 29:927–940. <https://doi.org/10.1016/j.eswa.2005.06.024>
- Fama EF (1970) Efficient capital markets: A review of theory and empirical work. *J Finance* 25:383–417. <https://doi.org/10.2307/2325486>
- Gal Y, Ghahramani Z (2015) Dropout as a bayesian approximation (eds) Representing model uncertainty in deep learning. Paper presented at the international conference on machine learning. <https://doi.org/10.48550/arXiv.1506.02142>
- Graves A, Schmidhuber J (2005) Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Netw* 18:602–610. <https://doi.org/10.1016/j.neunet.2005.06.042>
- Guo T, Xu Z, Yao X, Chen H, Aberer K, Funaya K (2016) Robust online time series prediction with recurrent neural networks (eds) IEEE International Conference on Data Science and Advanced Analytics (DSAA). <https://doi.org/10.1109/DSAA.2016.92>
- Guo Z, Wang H, Liu Q, Yang J (2014) A feature fusion based forecasting model for financial time series. *PLoS ONE* 9:e0101113. <https://doi.org/10.1371/journal.pone.0101113>
- Gurrib I (2018) Performance of the average directional index as a market timing tool for the most actively traded USD based currency pairs. *Banks Bank Syst* 13:58–70. [https://doi.org/10.21511/bbs.13\(3\).2018.06](https://doi.org/10.21511/bbs.13(3).2018.06)
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Comput* 9:1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hornik K, Stinchcombe M, White H (1989) Multilayer feedforward networks are universal approximators. *Neural Netw* 2:359–366. [https://doi.org/10.1016/0893-6080\(89\)90020-8](https://doi.org/10.1016/0893-6080(89)90020-8)
- Hoseinzade E, Haratizadeh S (2019) CNNpred: CNN-based stock market prediction using a diverse set of variables. *Expert Syst Appl* 129:273–285. <https://doi.org/10.1016/j.eswa.2019.03.029>
- Huang W, Nakamori Y, Wang S-Y (2005) Forecasting stock market movement direction with support vector machine. *Comput Oper Res* 32:2513–2522. <https://doi.org/10.1016/j.cor.2004.03.016>
- Jain S, Gupta R, Moghe AA (2018) Stock price prediction on daily stock data using deep neural networks (eds) International conference on advanced computation and telecommunication (ICACAT). <https://doi.org/10.1109/ICACAT.2018.8933791>
- Jiang M, Liu J, Zhang L, Liu C (2020) An improved stacking framework for stock index prediction by leveraging tree-based ensemble models and deep learning algorithms. *Physica A* 541:122272. <https://doi.org/10.1016/j.physa.2019.122272>
- Jiang W (2021) Applications of deep learning in stock market prediction: recent progress. *Expert Syst Appl* 184:115537. <https://doi.org/10.1016/j.eswa.2021.115537>

- Khan AH, Cao X, Li S, Katsikis VN, Liao L (2020) BAS-ADAM: an ADAM based approach to improve the performance of beetle antennae search optimizer. *IEEE/CAA J Automatica Sinica* 7:461–471. <https://doi.org/10.1109/JAS.2020.1003048>
- Kim H-j, Shin K-s (2007) A hybrid approach based on neural networks and genetic algorithms for detecting Temporal patterns in stock markets. *Appl Soft Comput* 7:569–576. <https://doi.org/10.1016/j.asoc.2006.03.004>
- Kingma DP, Ba J (2014) Adam: A method for stochastic optimization. *ArXiv Preprint*. <https://doi.org/10.48550/arXiv.1412.6980>. arXiv:1412.6980
- Komer B, Bergstra J, Eliasmith C (2019) Hyperopt-sklearn of chapter. In: (ed) *Automated machine learning*. Springer, Cham, City, pp 97–111
- Long W, Lu Z, Cui L (2019) Deep learning-based feature engineering for stock price movement prediction. *Knowl Based Syst* 164:163–173. <https://doi.org/10.1016/j.knosys.2018.10.034>
- Lu C-J, Lee T-S, Chiu C-C (2009) Financial time series forecasting using independent component analysis and support vector regression. *Decis Support Syst* 47:115–125. <https://doi.org/10.1016/j.dss.2009.02.001>
- Lu W, Li J, Wang J, Qin L (2021) A CNN-BiLSTM-AM method for stock price prediction. *Neural Comput Appl* 33:4741–4753. <https://doi.org/10.1007/s00521-020-05532-z>
- Maggipinto M, Masiero C, Beghi A, Susto GA (2018) A convolutional autoencoder approach for feature extraction in virtual metrology. *Procedia Manuf* 17:126–133. <https://doi.org/10.1016/j.promfg.2018.10.023>
- Mateus BC, Mendes M, Farinha JT, Assis R, Cardoso AM (2021) Comparing LSTM and GRU models to predict the condition of a pulp paper press. *Energies* 14:6958. <https://doi.org/10.3390/e14216958>
- Mostafa MM (2010) Forecasting stock exchange movements using neural networks: empirical evidence from Kuwait. *Expert Syst Appl* 37:6302–6309. <https://doi.org/10.1016/j.eswa.2010.02.091>
- Mukherjee S, Sadhukhan B, Sarkar N, Roy D, De S (2023) Stock market prediction using deep learning algorithms. *CAAI Trans Intell Technol* 8:82–94. <https://doi.org/10.1049/cit2.12059>
- Nabipour M, Nayyeri P, Jabani H, Mosavi A, Salwana E (2020) Deep learning for stock market prediction. *Entropy* 22:840. <https://doi.org/10.3390/e22080840>
- Nobre J, Neves RF (2019) Combining principal component analysis, discrete wavelet transform and XGBoost to trade in the financial markets. *Expert Syst Appl* 125:181–194. <https://doi.org/10.1016/j.eswa.2019.01.083>
- Papanicolaou A, Fu H, Krishnamurthy P, Healy B, Khorrami F (2023) An optimal control strategy for execution of large stock orders using long short-term memory networks. *J Comput Finance* 26. <https://doi.org/10.21314/JCF.2023.003>
- Patel M (2010) *Trading with Ichimoku clouds: the essential guide to Ichimoku Kinko Hyo technical analysis*. Wiley
- Ravi V, Pradeepkumar D, Deb K (2017) Financial time series prediction using hybrids of chaos theory, multi-layer perceptron and multi-objective evolutionary algorithms. *Swarm Evol Comput* 36:136–149. <https://doi.org/10.1016/j.swevo.2017.05.003>
- Savargiv M, Masoumi B, Keyvanpour MR (2024) An enhanced ensemble learning method for sentiment analysis based on Q-learning. *Iran J Sci Technol Trans Electr Eng* 1–17. <https://doi.org/10.1007/s40998-024-00718-w>
- Shynkevich Y, McGinnity TM, Coleman SA, Belatreche A, Li Y (2017) Forecasting price movements using technical indicators: investigating the impact of varying input window length. *Neurocomputing* 264:71–88. <https://doi.org/10.1016/j.neucom.2016.11.095>
- Siami-Namini S, Tavakoli N, Namin AS (2019) The performance of LSTM and BiLSTM in forecasting time series (eds) *IEEE International Conference on Big Data (Big Data)*. <https://doi.org/10.1109/BigData47090.2019.9005997>
- Spoorthy G, Sanjeevi S (2023) Multi-criteria–recommendations using autoencoder and deep neural networks with weight optimization using firefly algorithm. *Int J Eng* 36:130–138. <https://doi.org/10.5829/ije.2023.36.01a.15>
- Virtanen I, Yli-Olli P (1987) Forecasting stock market prices in a thin security market. *Omega* 15:145–155. [https://doi.org/10.1016/0305-0483\(87\)90029-6](https://doi.org/10.1016/0305-0483(87)90029-6)
- Wang C, Chen Y, Zhang S, Zhang Q (2022) Stock market index prediction using deep transformer model. *Expert Syst Appl* 208:118128. <https://doi.org/10.1016/j.eswa.2022.118128>
- Wang J-Z, Wang J-J, Zhang Z-G, Guo S-P (2011) Forecasting stock indices with back propagation neural network. *Expert Syst Appl* 38:14346–14355. <https://doi.org/10.1016/j.eswa.2011.04.222>
- Wiese M, Knobloch R, Korn R, Kretschmer P (2020) Quant gans: deep generation of financial time series. *Quant Finance* 20:1419–1440. <https://doi.org/10.1080/14697688.2020.1730426>
- Xia Y, Liu C, Li Y, Liu N (2017) A boosted decision tree approach using bayesian hyper-parameter optimization for credit scoring. *Expert Syst Appl* 78:225–241. <https://doi.org/10.1016/j.eswa.2017.02.017>
- Xu Y, Zhang Y, Liu P, Zhang Q, Zuo Y (2024) GAN-Enhanced non-linear fusion model for stock price prediction. *Int J Comput Intell Syst* 17:12. <https://doi.org/10.1007/s44196-023-00394-4>
- Yang F, Chen Z, Li J, Tang L (2019) A novel hybrid stock selection method with stock prediction. *Appl Soft Comput* 80:820–831. <https://doi.org/10.1016/j.asoc.2019.03.028>
- Yang M, Wang J (2022) Adaptability of financial time series prediction based on BiLSTM. *Procedia Comput Sci* 199:18–25. <https://doi.org/10.1016/j.procs.2022.01.003>
- Yang Z, Zhao T, Wang S, Li X (2024) MDF-DMC: A stock prediction model combining multi-view stock data features with dynamic market correlation information. *Expert Syst Appl* 238:122134. <https://doi.org/10.1016/j.eswa.2023.122134>
- Yi K, Zhang Q, Fan W, Cao L, Wang S, He H, Xiong H (2025) A survey on deep learning based time series analysis with frequency transformation (eds) *Paper presented at the Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*
- Zare Mehrjardi Y, Jasemi M, Abooei MH, Ahmadi E (2016) A nonlinear autoregressive model with exogenous variables neural network for stock market timing: the candlestick technical analysis. *Int J Eng* 29:1717–1725. <https://doi.org/10.5829/idosi.ije.2016.29.12c.10>
- Zhou H, Zhang S, Peng J, Zhang S, Li J, Xiong H, Zhang W (2021) Informer: Beyond efficient transformer for long sequence time-series forecasting. *Paper presented at the Proceedings of the AAAI conference on artificial intelligence*
- Zięba M, Tomczak SK, Tomczak JM (2016) Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction. *Expert Syst Appl* 58:93–101. <https://doi.org/10.1016/j.eswa.2016.04.001>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.